

# Noise Robust Radar HRRP Target Recognition Based on Scatterer Matching Algorithm

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**Abstract**—Since the signal-to-noise ratio (SNR) directly relates to the distance between the target and the radar for a given noise power and radar power, the noise robustness of a recognition algorithm is very important to increase the recognition distance between the target and the radar in the real application. In this paper, a novel noise-robust recognition method for high-resolution range profile (HRRP) data is proposed to enhance its recognition performance under the test condition of low SNR. The target dominant scatterers are first extracted based on the scattering center model of complex HRRP data via the orthogonal matching pursuit algorithm. Then, a scatterer matching recognition algorithm based on Hausdorff distance is developed with the magnitudes and locations of extracted dominant scatterers used as the feature patterns. Here, the noise reduction is accomplished based on the sparse distribution property of dominant scattering centers in a target. Experimental results on the synthetic and measured HRRP data demonstrate that the proposed method can improve the recognition performance under the relatively low SNR condition for both orthogonal and superresolution representations of scattering center model.

**Index Terms**—Radar automatic target recognition (RATR), high-resolution range profile (HRRP), orthogonal matching pursuit (OMP), point pattern matching, Hausdorff distance.

## I. INTRODUCTION

**R**ADAR high-resolution range profile (HRRP) has received intensive attention from the radar automatic target recognition (RATR) community [1], [2], [4]–[6], [9], [14], [17]–[19], [21], [29], [30]. For the RATR problem, the training data are usually collected under the condition of high signal-to-noise ratio (SNR) via some cooperative measurement experiments or directly via simulations; while the test samples are usually got in the non-cooperative circumstance (e.g., at the battle time), where the high SNR condition cannot be guaranteed due to the long distance away from the non-cooperative targets [7], [8], [12], [15], [24]. As shown in the experimental results in [6], the recognition performance dramatically deteriorates under the low test SNR. Therefore, the

noise-robustness of a recognition algorithm is very important in the real application.

There are usually three approaches to improve the recognition performance of HRRP data under the low test SNR.

- One is to extract the noise-robust features from the noised HRRP test samples. As discussed in [15] and [24], the bispectrum feature of the complex radar signal can suppress additive disturbances that are described by the symmetric probability density functions (pdfs), e.g., additive white Gaussian noise, additive colored Gaussian noise, and additive non-Gaussian noise. Although the bispectrum features proposed in [15] and [24] are robust to the variance of noise level, and can yield the good recognition performance under the relatively low test SNR, both methods require a HRRP sequence to get the ensemble average estimation for the bispectrum feature. Such a requirement in the test stage will definitely decrease the recognition speed.
- Several literatures [2], [4], [6], [9], [14], [18], [19], [21] showed that statistical recognition is an efficient method for HRRP based RATR. By statistical recognition is meant that the class membership of a measurement HRRP sample  $\mathbf{x}$  is determined based on the posterior probabilities of all class memberships  $p(g|\mathbf{x})$  ( $g \in \{1, 2, \dots, G\}$  denotes class membership). According to the Bayes' theorem, we have  $p(g|\mathbf{x}) \propto p(\mathbf{x}|g)p(g)$ , where  $p(\mathbf{x}|g)$  is the class-conditional likelihood and  $p(g)$  is the prior class probability. Since the prior class probability is usually assumed to be uniformly distributed, estimation of the posterior probability  $p(g|\mathbf{x})$  turns into estimation of the class-conditional likelihood  $p(\mathbf{x}|g)$ . In order to get the robust recognition performance via a statistical recognition method, the statistical model learned in the training stage,  $p(\mathbf{x}|g)$ , should be modified to match the variation of the SNR condition in the classification stage. For the radar signal processing, the noise signal directly takes effects on the complex-valued target echo, which can be expressed as a linear model, i.e.,  $\tilde{\mathbf{x}} = \mathbf{x} + \mathbf{v}$  with  $\tilde{\mathbf{x}}$  denoting the complex-valued noised signal,  $\mathbf{x}$  denoting the complex-valued signal without noise or under high SNR, and  $\mathbf{v}$  denoting the complex-valued noise variable introduced from the test condition. Based on the learned  $p(\mathbf{x}|g)$  and such a linear structure, the statistical model  $p(\tilde{\mathbf{x}}|g)$  can be easily and accurately derived in the complex domain. However, the absolute value operator to obtain the real-valued HRRP feature completely changes the linear structure, i.e.,  $|\tilde{\mathbf{x}}| = |\mathbf{x} + \mathbf{v}| \neq |\mathbf{x}| + |\mathbf{v}|$ . Thus it

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is difficult to accurately get the update statistical model in the real-valued domain, *i.e.*, calculating  $p((|\bar{x}|)|g)$  from  $p((|x|)|g)$ , to match the variation of SNR. Based on the additive real-valued noise assumption, [7] proposes a noise-robust modification method for multitask-learning based factor analysis model  $p((|x|)|g)$  of real-valued HRRP data  $|x|$ . Although the good noise robustness is achieved via the proposed method, it suffers from a high computational burden to adaptively modify the parameters related to the SNR level in the statistical model.

- The last but also the most natural choice is to remove the noise component in radar echo before classification. Wavelet shrinkage is proposed as a denoising method for HRRP data in [12]. As shown in the experiments in [12], its performance depends on the approximation-detail threshold setting and degrades largely in the presence of severe noise contamination.

The research reported here seeks a denoising method based on sparse representation of HRRP data and develops a scatterer matching recognition algorithm for the denoised data. As discussed in [28], based on the scattering center model, the dominant scattering centers sparsely spread over the Fourier dictionary, of which the echoes can approximate the target signal; while the weak scattering centers spread uniformly over the Fourier dictionary, which correspond to the noise components. According to the sparse representation theory, such a target signal can be stably recovered from the noisy measurements [11]. Here the scattering coefficients and locations of the dominant scattering centers are first estimated for a noisy test sample by solving the sparse optimization problem with the noise level constraint, then the scatterer matching algorithm based on Hausdorff distances (HDs) between its dominant scatterers and those from the training templates is used to distinguish the test sample. Thus the denoised sample doesn't need to be recovered in our noise-robust recognition method, while the magnitudes and locations of the extracted dominant scatterers can be directly employed for our recognition algorithm. In the experiments based on measured HRRP dataset, the proposed method shows the inspiring recognition performance under the relatively low SNR condition for both orthogonal and superresolution Fourier dictionaries based sparse representations.

The remainder of the paper is organized as follows. We introduce the scattering center model and denoising method in Section II. The scatterer matching recognition method is developed in Section III-B. The detailed experimental results based on measured HRRP data are provided in Section IV, followed by conclusions in Section V.

## II. DENOISING METHOD BASED ON SPARSE REPRESENTATION

### A. Scattering Center Model for Complex HRRP Data

Here the transmitted signal  $x_t(t)$  is the linear frequency modulated (LFM) signal.

$$x_t(t) = \text{rect}\left(\frac{t}{T_p}\right) \exp(j2\pi f_c t + j\pi \mu t^2) \quad (1)$$

where  $\text{rect}(\frac{t}{T_p})$  denotes a rectangular pulse of width  $T_p$ ,  $T_p$  denotes the pulse width,  $f_c$  denotes the radar center frequency,  $\mu$  denotes the LFM coefficient with the bandwidth  $B = \mu T_p$ . According to [16], the high-frequency scattering response of an object is well approximated as a sum of responses from individual scattering centers [16]. Thus the complex returned echo in baseband is denoted as

$$x_r(t) = \sum_{n=1}^{K_0} A_n \text{rect}\left(\frac{t - \frac{2R_n}{c}}{T_p}\right) \times \exp\left(-j2\pi f_c \frac{2R_n}{c} + j\pi \mu \left(t - \frac{2R_n}{c}\right)^2\right) \quad (2)$$

where  $R_n$  denotes the radial distance between the  $n$ th scatterer and the radar,  $A_n$  denotes the corresponding scattering coefficient,  $K_0$  is the number of scatterers,  $c$  denotes the propagation velocity of radar signal. Assume that  $u(t) = \text{rect}(\frac{t}{T_p}) \exp(j\pi \mu t^2)$  denotes the envelope of the transmitted signal and  $U(f)$  represents its frequency response, then the Fourier transformation of (2) is

$$y_r(f) = \sum_{n=1}^{K_0} A_n U(f) \exp\left(-j2\pi (f_c + f) \frac{2R_n}{c}\right) \quad (3)$$

By pulse compression technique, we get

$$y(f) = y_r(f) \cdot U^*(f) = \sum_{n=1}^{K_0} A_n P(f) \exp\left(-j2\pi (f_c + f) \frac{2R_n}{c}\right) \quad (4)$$

where  $U^*(f)$  denotes the conjugate form of  $U(f)$ ,  $P(f)$  is the power spectrum of  $u(t)$ . For a LFM signal,  $P(f) \approx \text{rect}(\frac{f}{B})$ . Then (4) can be written as

$$y(f) = \sum_{n=1}^{K_0} A_n \exp(-j2\pi f_c \frac{2R_n}{c}) \cdot \exp(-j2\pi f \frac{2R_n}{c}) \\ = \sum_{n=1}^{K_0} \omega(n) \cdot \exp(-j2\pi f \frac{2R_n}{c}), \quad -\frac{B}{2} < f < \frac{B}{2} \quad (5)$$

where  $\omega(n) = A_n \exp(-j2\pi f_c \frac{2R_n}{c})$ . The discrete frequency response can be represented by

$$y(l) = \sum_{n=1}^{K_0} \omega(n) \cdot \exp\left(-j2\pi \frac{2R_n}{c} (l-1)\Delta f\right) \quad (6)$$

where  $y(l)$  represents the  $l$ th frequency response with  $l \in \{1, \dots, L\}$  and  $L$  denoting the number of frequency components,  $\Delta f$  denotes the frequency interval between the neighboring frequency components in the discrete frequency response. In (6), the parameters to be estimated are  $\{K_0, \{\omega(n), R_n\}_{n=1}^{K_0}\}$ .

For a wideband radar system, the range dimension is divided into the range cells by its range resolution  $\Delta R = \frac{c}{2B}$ . If we assume that each  $R_n$  in (6) is an integer multiple of  $\Delta R$ ,

then (6) can be rewritten as

$$\begin{aligned} y(l) &= \sum_{k=1}^{K, r_k \in \{R_n\}_{n=1}^{K_0}} w(k) \cdot \exp\left(-j2\pi \frac{2r_k}{c}(l-1)\Delta f\right) \\ &\quad + \sum_{k=1}^{K, r_k \notin \{R_n\}_{n=1}^{K_0}} w(k) \cdot \exp\left(-j2\pi \frac{2r_k}{c}(l-1)\Delta f\right) \\ &= \sum_{k=1}^K w(k) \cdot \exp\left(-j2\pi \frac{2r_k}{c}(l-1)\Delta f\right) \end{aligned} \quad (7)$$

where  $\{r_k\}_{k=1}^K$  denote the locations of range cells with  $k \in \{1, \dots, K\}$  and  $K > K_0$  denoting the number of range cells,  $w(k)$  represents the corresponding scattering coefficient from the  $k$ th range cell. For  $r_k = R_{n'}$  with  $R_{n'} \in \{R_n\}_{n=1}^{K_0}$ ,  $w(k) = \omega(n')$ ; otherwise,  $w(k) = 0$ . In this expression (7),  $K_0$  equals to the number of the non-zero elements in vector  $\mathbf{w}$ , and  $r_k = k \cdot \Delta R$ , then the parameters to be estimated are only  $\{w(k)\}_{k=1}^K$ .

Let

$$\begin{aligned} \phi(r_k) &= \left[ 1, \exp(-j\frac{4\pi}{c}r_k\Delta f), \exp(-j\frac{4\pi}{c}r_k2\Delta f), \dots, \exp(-j\frac{4\pi}{c}r_k(L-1)\Delta f) \right]^T \\ \Phi &= [\phi(r_1), \phi(r_2), \dots, \phi(r_K)] \end{aligned} \quad (8)$$

Considering the noise component in the real application, the signal model in (7) can be expressed in a vector-matrix form as

$$\mathbf{y} = \Phi \mathbf{w} + \boldsymbol{\eta} \quad (9)$$

where  $\mathbf{y} = [y(1), y(2), \dots, y(L)]^T$ ,  $\boldsymbol{\eta} = [\eta(1), \eta(2), \dots, \eta(L)]^T$  represents the noise component,  $\mathbf{w} = [w(1), w(2), \dots, w(K)]^T$  denotes the complex HRRP sample, and  $|\mathbf{w}|$  is the corresponding real HRRP sample. When  $K = L$ ,  $\Phi$  is a complete orthogonal basis matrix; when  $K > L$ ,  $\Phi$  is a non-orthogonal and redundant dictionary matrix, then the superresolution representation can be achieved.

### B. Noise Reduction via Orthogonal Matching Pursuit Algorithm

According to the scattering center model introduced above, the denoised frequency response can be recovered as  $\hat{\mathbf{y}} = \Phi \hat{\mathbf{w}}$ , where  $\hat{\mathbf{w}}$  denotes the estimated scattering coefficients from the target scatterers. As discussed in [28] for CS-based ISAR imaging, the target signal can be approximated by the echoes from the dominant scattering centers which are assumed to be sparsely distributed in the target, while those from the weak scattering centers are regarded as the noise components. Therefore,  $\hat{\mathbf{w}}$  can be estimated by solving the following sparse optimization problem

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \|\mathbf{y} - \Phi \mathbf{w}\|_2^2 \quad \text{s.t.} \quad \|\mathbf{w}\|_0 = K_0 \quad (10)$$

where  $\|\cdot\|_2$  denotes the  $l_2$  norm,  $\|\cdot\|_0$  denotes the  $l_0$  norm and  $\|\mathbf{w}\|_0 = K_0$  represents that there are  $K_0$  non-zero elements in vector  $\mathbf{w}$ .

According to the sparse representation theory [27], if  $K_0 < (M^{-1} + 1)/2$ , it is a unique solution of (10). Here  $M$  denotes

the maximum correlation between the columns in the basis matrix  $\Phi$ , that is

$$M = \arg \max_{k_1 \neq k_2} |\phi(r_{k_1})^H \phi(r_{k_2})| \quad (11)$$

where  $(\cdot)^H$  is conjugate transpose. Solving (10) requires a combinatorial search and is generally NP-hard [3]. If  $K_0$  is known as the prior information, the orthogonal matching pursuit (OMP) algorithm [11] can be utilized for (10) to get its approximate solution. OMP is an iterative greedy algorithm that selects at each step the column of  $\Phi$  which is most correlated with the current residuals. This column is then added into the set of selected columns. The algorithm updates the residuals by projecting the observation  $\mathbf{y}$  onto the linear subspace spanned by the columns that have already been selected and the algorithm then iterates. After the  $K_0$ th iteration, the  $K_0$  target scatterers can be got, and the residual component is from noise. The denoised HRRP sample can be obtained by setting  $w(k) = 0$  for the  $k$ th scatterer unselected as one of the  $K_0$  target scatterers. Compared with other alternative methods for (10), a major advantage of the OMP is its simplicity and fast implementation.

In real application, it is hard to know the number of target scatterers  $K_0$ , but the noise power  $\delta$  in the frequency response from the noised HRRP sample can be estimated via some pre-processing methods. The noise considered here mainly refers to the heat noise from the receiver, which can be assumed to be stationary, accordingly, there are several ways to accurately estimate the noise power [23], of which a simple approach is to measure the noise power from the receiver with no input signal. Thus it is reasonable to terminate the OMP algorithm when the residual signal power is lower than the noise power. In this way, the sparse optimization problem should be

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \|\mathbf{w}\|_0 \quad \text{s.t.} \quad \|\mathbf{y} - \Phi \mathbf{w}\|_2^2 < \delta \quad (12)$$

where  $\delta$  denotes the estimated noise power in the frequency response,  $|\hat{\mathbf{w}}|$  is the denoised HRRP sample. The OMP algorithm for target scatterer extraction is summarized in Algorithm 1.

### III. SCATTERER MATCHING RECOGNITION METHOD

As discussed in Section II, the target scattering coefficient vector  $\hat{\mathbf{w}}$  can be learned via the OMP algorithm for a complex HRRP sample. The indices and absolute values of the nonzero entries in  $\hat{\mathbf{w}}$  indicate the locations and magnitudes of the target scatterers. In the following, we will develop a scatterer matching algorithm with the magnitudes and locations of extracted dominant scatterers used as the feature patterns.

#### A. Hausdorff Distance Between Point Sets

The Hausdorff measure is a well-known method for representing the distance between two point sets without having an a priori correspondence between the two sets [13]. Given the two point sets  $\mathbb{A} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_N\}$  and  $\mathbb{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_Q\}$ , the Hausdorff distance (HD) is

**Algorithm 1** OMP for Target Scatterer Extraction

**Input:** The noisy frequency response  $\mathbf{y}$ , the estimated noise power  $\delta$ , and the basis matrix  $\Phi$ .

**Initialization:** The initial set of target scatterers  $\mathbb{S}^0 = \emptyset$ , the residual signal  $\epsilon^0 = \mathbf{y}$  and the iteration counter  $t = 0$ .

**repeat**

- $t = t + 1$ .
- Calculate  $e(k) = \min_{z_k} \|\phi(r_k)z_k - \epsilon^{t-1}\|_2^2$ , where  $\phi(r_k)$  denotes the  $k$ th column of  $\Phi$ , and the least squares solution for  $z_k$  is  $z_k = \phi(r_k)^H \epsilon^{t-1}$ .
- Find  $\tilde{k} = \arg \min_k e(k)$ , and let the new set of target scatterers  $\mathbb{S}^t = \mathbb{S}^{t-1} \cup \{\tilde{k}\}$ .
- Update the scattering coefficients  $\hat{\mathbf{w}}$  with the nonzero entries are located at the indices indicated by  $\mathbb{S}^t$  with  $\hat{\mathbf{w}}_{\mathbb{S}^t}^t = \arg \min_{\mathbf{w}} \|\mathbf{y} - \Phi(\mathbb{S}^t)\mathbf{w}\|_2^2$ , where the basis matrix  $\Phi(\mathbb{S}^t)$  is composed of the column vectors  $\phi(r_k)$  with  $k \in \mathbb{S}^t$ . The least squares solution for  $\hat{\mathbf{w}}_{\mathbb{S}^t}^t$  is  $\hat{\mathbf{w}}_{\mathbb{S}^t}^t = [\Phi_{\mathbb{S}^t}^H \Phi_{\mathbb{S}^t}]^{-1} \Phi_{\mathbb{S}^t}^H \mathbf{y}$ .
- Update the residual signal  $\epsilon^t = \mathbf{y} - \Phi \hat{\mathbf{w}}$ .

**until**  $\|\epsilon^t\|_2^2 \leq \delta$ .

**Output:** The target scattering coefficients  $\hat{\mathbf{w}}$ .

defined as

$$H(\mathbb{A}, \mathbb{B}) = \max \left( h(\mathbb{A}, \mathbb{B}), h(\mathbb{B}, \mathbb{A}) \right) \quad (13)$$

with

$$h(\mathbb{A}, \mathbb{B}) = \max_{\mathbf{a}_n \in \mathbb{A}} \min_{\mathbf{b}_q \in \mathbb{B}} d(\mathbf{a}_n, \mathbf{b}_q) \quad (14)$$

where the function  $h(\mathbb{A}, \mathbb{B})$  is called the directed Hausdorff distance (DHD), and  $d(\mathbf{a}_n, \mathbf{b}_q)$  denotes a distance between pattern vectors  $\mathbf{a}_n$  and  $\mathbf{b}_q$ . The DHD is the distance of the most isolated point between  $\mathbb{A}$  and  $\mathbb{B}$ . However, an outlier or occlusion could skew an otherwise close correspondence. To solve this problem, some partial Hausdorff distance (PHD) measures are proposed, of which the least trimmed square Hausdorff distance (LTS-HD) is regarded to be robust [10], [22]. The directed LTS-HD may be written

$$h_{\text{LTS}}(\mathbb{A}, \mathbb{B}) = \frac{1}{P} \sum_{\mathbf{a}_{n_1} \in \mathbb{A}'} \min_{\mathbf{b}_q \in \mathbb{B}} d(\mathbf{a}_{n_1}, \mathbf{b}_q) \quad (15)$$

where  $\mathbb{A}' = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_P\}$ ,  $\mathbb{A}' \subset \mathbb{A}$ ,  $P < N$ ,  $\min_{\mathbf{b}_q \in \mathbb{B}} d(\mathbf{a}_{n_1}, \mathbf{b}_q) < \min_{\mathbf{b}_q \in \mathbb{B}} d(\mathbf{a}_{n_2}, \mathbf{b}_q)$  for each  $\mathbf{a}_{n_1} \in \mathbb{A}'$  and  $\mathbf{a}_{n_2} \in \mathbb{A} - \mathbb{A}'$ . Thus LTS-HD takes the mean of the  $P$  minimum distances between the point sets. Paper [20] suggests setting  $P$  in (15) as a fraction of the cardinality of one of the sets, e.g.,  $P = \lfloor r \min\{N, Q\} \rfloor$  with  $0 < r \leq 1$  and  $\lfloor m \rfloor$  denoting to round the number  $m$  to the nearest integers towards minus infinity.

Since scatterer distribution is the important information contained in HRRP data, the locations and magnitudes of the target scatterers can be used as the feature patterns for HRRP target recognition. Let  $\mathbb{S}_{\mathbb{A}} = \{s_{\mathbb{A}1}, s_{\mathbb{A}2}, \dots, s_{\mathbb{A}N}\}$  denote the location indices of scatterers extracted from an HRRP sample and  $|\hat{\mathbf{w}}_{\mathbb{S}_{\mathbb{A}}}| = [|\hat{w}_{s_{\mathbb{A}1}}|, |\hat{w}_{s_{\mathbb{A}2}}|, \dots, |\hat{w}_{s_{\mathbb{A}N}}|]^T$  denote the corresponding magnitudes, which are extracted via the OMP algorithm discussed in Section II-B, the feature pattern for each scatterer is  $\mathbf{a}_n = [s_{\mathbb{A}n}, |\hat{w}_{s_{\mathbb{A}n}}|]^T$  and the point set for this HRRP sample is  $\mathbb{A} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_N\}$ , where the pattern order, i.e., the scatterers' order, in  $\mathbb{A}$  accord with their extraction order via the OMP algorithm. According to Algorithm 1, the scatterers' magnitudes must satisfy  $|\hat{w}_{s_{\mathbb{A}1}}| \geq |\hat{w}_{s_{\mathbb{A}2}}| \geq \dots \geq |\hat{w}_{s_{\mathbb{A}N}}|$ . Thus the first extracted

scatterers are dominant scatterers for an HRRP sample. As discussed in [28], the target signal can be approximated by the echoes from the dominant scattering centers. Thus we propose a PHD measure, called dominant-scatterers' Hausdorff distance (ds-HD), for our scatterer matching problem. The ds-HD is expressed as

$$\begin{aligned} h_{\text{ds}}(\mathbb{A}, \mathbb{B}) &= \frac{1}{P_1} \sum_{\mathbf{a}_{n_1} \in \mathbb{A}'} \min_{\mathbf{b}_{q_1} \in \mathbb{B}'} d(\mathbf{a}_{n_1}, \mathbf{b}_{q_1}), \\ h_{\text{ds}}(\mathbb{B}, \mathbb{A}) &= \frac{1}{P_2} \sum_{\mathbf{b}_{q_1} \in \mathbb{B}'} \min_{\mathbf{a}_{n_1} \in \mathbb{A}'} d(\mathbf{b}_{q_1}, \mathbf{a}_{n_1}) \end{aligned} \quad (16)$$

In (16),  $\mathbb{A}' = \mathbb{A}'_1 \cup \{\mathbf{a}_{P_1}\} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{P_1-1}\} \cup \{\mathbf{a}_{P_1}\}$ ,  $\mathbb{A}' \subset \mathbb{A}$ ,  $P_1 < N$ ,  $\|\Phi \hat{\mathbf{w}}_{\mathbb{S}_{\mathbb{A}'}}\|_2 < r \|\mathbf{y}_{\mathbb{A}}\|_2$  and  $\|\Phi \hat{\mathbf{w}}_{\mathbb{S}_{\mathbb{A}'}}\|_2 \geq r \|\mathbf{y}_{\mathbb{A}}\|_2$  with  $\mathbf{y}_{\mathbb{A}}$  denoting the frequency response of the HRRP sample corresponding to  $\mathbb{A}$  and  $0 < r < 1$ . Thus  $\mathbb{A}'$  represents the dominant scatterers' pattern set of the HRRP sample. Similarly for  $\mathbb{B}'$ ,  $\mathbb{B}' = \mathbb{B}'_1 \cup \{\mathbf{b}_{P_2}\} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_{P_2-1}\} \cup \{\mathbf{b}_{P_2}\}$ ,  $\mathbb{B}' \subset \mathbb{B}$ ,  $P_2 < Q$ ,  $\|\Phi \hat{\mathbf{w}}_{\mathbb{S}_{\mathbb{B}'}}\|_2 < r \|\mathbf{y}_{\mathbb{B}}\|_2$  and  $\|\Phi \hat{\mathbf{w}}_{\mathbb{S}_{\mathbb{B}'}}\|_2 \geq r \|\mathbf{y}_{\mathbb{B}}\|_2$  with  $\mathbf{y}_{\mathbb{B}}$  denoting the frequency response of the HRRP sample corresponding to  $\mathbb{B}$  and  $0 < r < 1$ . Therefore, the ds-HD is the mean of distances between the dominant scatterers from two point sets. The PHD used in this paper is defined as

$$H(\mathbb{A}, \mathbb{B}) = \max \left( h_{\text{ds}}(\mathbb{A}, \mathbb{B}), h_{\text{ds}}(\mathbb{B}, \mathbb{A}) \right) \quad (17)$$

The  $d(\cdot, \cdot)$  in (16) is calculated from the Mahalanobis distance

$$d(\mathbf{a}_{n_1}, \mathbf{b}_{q_1}) = \sqrt{(\mathbf{a}_{n_1} - \mathbf{b}_{q_1})^T \Sigma^{-1} (\mathbf{a}_{n_1} - \mathbf{b}_{q_1})} \quad (18)$$

where  $\Sigma$  is a diagonal matrix, its diagonal entries are the measurement error variances of each feature in the feature vectors in the point sets. The Mahalanobis distance facilitates the comparison of vectors, where the various dimensions have different scales. Here one feature in our pattern is the location index of a scatterer, while the other is the intensity of the scatterer. Their scales are different.

### B. HRRP Recognition Algorithm Based on Scatterer Matching

As discussed in [2], [4]–[7], [9], [14], [18], [19], and [21], HRRPs from complex targets yield target signatures that are



**Algorithm 2** Training Stage of the Scatterer Matching Recognition Algorithm

**Input:** The training frequency response data  $\{\mathbf{y}_d^{(g,m)}\}_{d=1,m=1,g=1}^{D,M_g,G}$  with  $D$  denoting the number of samples in each aspect-frame,  $M_g$  denoting the number of aspect-frames of the  $g$ th target and  $G$  denoting the number of targets.

**For**  $g = 1$  **to**  $G$

**For**  $m = 1$  **to**  $M_g$

- Time-shift compensate and amplitude normalize each training sample  $\mathbf{y}_d^{(g,m)}$  with its first geometric moment and  $l_2$ -norm.
- Extract dominant scatterers from the time-shift compensated and amplitude normalized training samples  $\{\mathbf{y}_d^{(g,m)}\}_{d=1}^D$  via Algorithm 1. The scatterer set  $\mathbb{A}_d^{(g,m)}$  is made up of the location index and intensity of each dominant scatterer extracted from the sample  $\mathbf{y}_d^{(g,m)}$ .
- Calculate PHD  $H(\mathbb{A}_d^{(g,m)}, \mathbb{A}_{d'}^{(g,m)})$  via (16), (17) and (18) for each sample in the aspect frame with  $d \neq d'$ .
- Find the scatterer set  $\mathbb{A}_{d_1}^{(g,m)}$  with the minimum summation of PHDs between  $\mathbb{A}_d^{(g,m)}$  and  $\mathbb{A}_{d_1}^{(g,m)}$  with  $d \neq d_1$  in this aspect frame.

$$\mathbb{A}_{d_1}^{(g,m)} = \arg \min_{\mathbb{A}_{d_1}^{(g,m)}} \sum_{d \neq d_1} H(\mathbb{A}_d^{(g,m)}, \mathbb{A}_{d_1}^{(g,m)})$$

**End**

**End**

**Output:** The frame template  $\{\mathbb{A}_{d_1}^{(g,m)}\}_{m=1,g=1}^{M_g,G}$ .

**Algorithm 3** Classification Stage of the Scatterer Matching Recognition Algorithm

**Input:** The test frequency response  $\mathbf{y}$ , the frame templates  $\{\mathbb{A}_{d_1}^{(g,m)}\}_{m=1,g=1}^{M_g,G}$  with  $M_g$  denoting the number of aspect-frames of the  $g$ th target and  $G$  denoting the number of targets.

- Time-shift compensate and amplitude normalize the test sample  $\mathbf{y}$  with its first geometric moment and  $l_2$ -norm.
- Extract dominant scatterers from the time-shift compensated and amplitude normalized test sample  $\mathbf{y}$  via Algorithm 1.
- Calculate PHD  $H(\mathbb{A}, \mathbb{A}_{d_1}^{(g,m)})$  via (16), (17) and (18) with  $m \in \{1, 2, \dots, M_g\}$  and  $g \in \{1, 2, \dots, G\}$ , where  $\mathbb{A}$  denotes the scatterer set corresponding to the test sample with the location index and intensity of each dominant scatterer used as the feature pattern.
- Find the scatterer set  $\mathbb{A}_{d_1}^{(g^*,m^*)}$  with the minimum PHD  $H(\mathbb{A}, \mathbb{A}_{d_1}^{(g^*,m^*)})$ , i.e.,  $\{g^*, m^*\} = \arg \min_{g,m} H(\mathbb{A}, \mathbb{A}_{d_1}^{(g,m)})$ .

**Output:** The class membership of the test sample  $g^*$ .

a strong function of the target-sensor orientation, referred to as target-aspect sensitivity in [4]–[6] and [21]. To deal with the target-aspect sensitivity, the multi-aspect HRRP dataset from each target is divided into subsets (as in [4]–[6], [25], and [26]), where the HRRP data in each subset are collected from a target-aspect sector roughly without the scatterers' motion through range cells (MTRC). Thus each subset is defined to be an aspect-frame from the target [4]–[6], [25]. Due to no MTRC, the HRRPs within such a small aspect-frame change little and their scatterer distribution structures can be assumed to be same. For each aspect-frame from the training data, we use the HRRP sample with the smallest summation of PHDs between it and each other samples in this aspect-frame as the frame template. In the classification stage, the denoising processing is implemented for a single test sample. Then according to the nearest neighbor criterion, the class membership of the test sample will be assigned to the class of the frame template with the minimum PHD.

Since an HRRP sample is only a part of the received radar echo extracted by a range window, in which target signal is included, the position of the target signal in an HRRP sample can vary with the measurement. This is the time-shift sensitivity. Thus the PHD cannot be calculated for two HRRP samples directly via the extracted scatterers without any time-shift compensation preprocessing. In this paper,

each training or test sample is time-shift compensated with its first geometric moment, which is also used as the time-shift compensation preprocessing method in [1], [18], and [21].

Furthermore, since the intensity of an HRRP sample is a function of radar transmitting power, target distance, radar antenna gain, radar receiver gain, radar system losses and so on, HRRP samples measured by different radars or under different conditions will have different amplitude-scales. To deal with this amplitude-scale sensitivity, each complex frequency response sample is normalized by dividing its  $l_2$ -norm.

The detailed training and classification stages of the scatterer matching recognition algorithm are summarized in Algorithms 2 and 3.

## IV. EXPERIMENTAL RESULTS

## A. Measured Data

We examine the performance of the recognition method on the measured HRRP data from the three real airplanes, which are also used in [4]–[7], [9], and [21]. The parameters of the airplane targets and measurement radar are shown in Table I, and the projections of target trajectories onto the ground plane are shown in Figure 1, from which the aspect angle of an airplane can be estimated according to

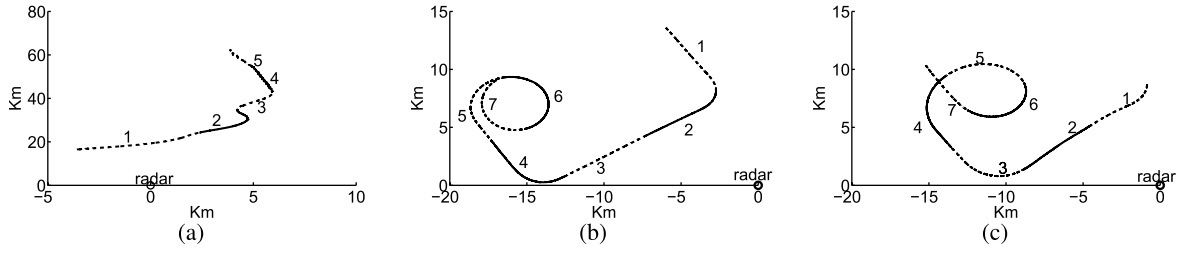


Fig. 1. Projections of target trajectories onto the ground plane: (a) Yak-42; (b) Cessna Citation S/II; and (c) An-26.

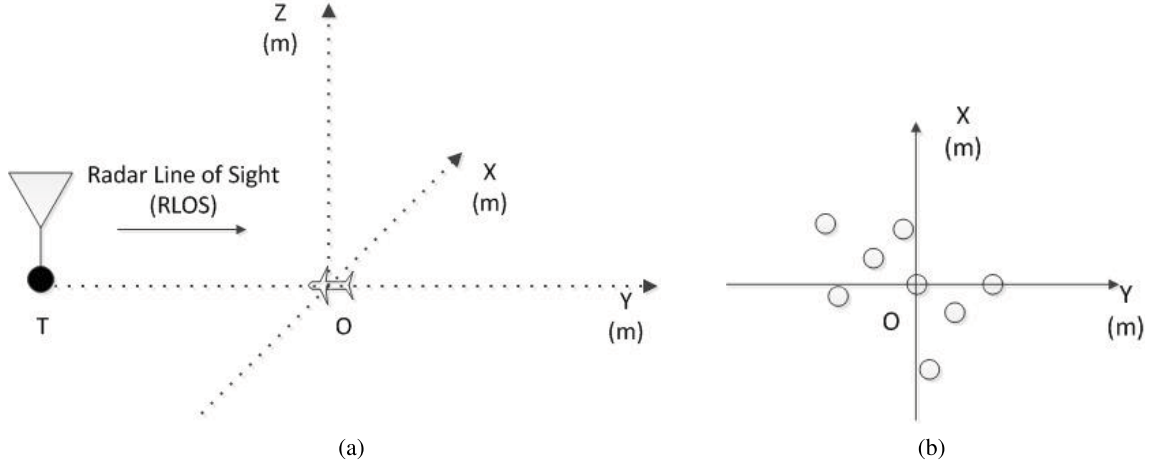


Fig. 2. Simulation example for a synthetic HRRP: (a) the positions of the radar and airplane target; and (b) the scatterer distribution of the airplane target.

TABLE I  
PARAMETERS OF PLANES AND RADAR IN THE  
MEASUREMENT EXPERIMENT

| radar parameters     | center frequency $f_0$ |           | 5520 MHz   |
|----------------------|------------------------|-----------|------------|
|                      | bandwidth $B$          |           | 400 MHz    |
| planes               | length (m)             | width (m) | height (m) |
| Yak-42               | 36.38                  | 34.88     | 9.83       |
| Cessna Citation S/II | 14.40                  | 15.90     | 4.57       |
| An-26                | 23.80                  | 29.20     | 9.83       |

its relative position to radar. As shown in Figure 1, the measured data are segmented, where each segment contains 25600 HRRP samples and each sample is a 256-length vector, *i.e.*,  $L = 256$ . In our recognition experiments, data from the 2nd segment and a part of the 5th segment of Yak-42, the 6th and the 7th segments of Cessna Citation S/II, the 5th and the 6th segments of An-26 are taken as the training samples, and other data segments are taken as test samples.

This HRRP dataset is measured under high SNR ( $\approx 40$ dB) and without any interference. In order to evaluate the denoising performance of our proposed method, we add simulated white noises to the inphase and quadrature components of the complex frequency responses from the test samples. The SNR is defined as

$$\text{SNR} = 10 \times \log_{10} \left( \frac{\sum_{l=1}^L P_{y_l}}{L \times P_{\text{Noise}}} \right) \quad (19)$$

where  $P_{y_l}$  denotes the power of the  $l$ th frequency response from the original test sample,  $L$  denotes the number of

frequency components (here  $L = 256$ ), and  $P_{\text{Noise}}$  denotes the power of noise in each frequency component.

### B. Denoising Performance

1) *Synthetic Data Reconstruction*: Firstly, we demonstrate the denoising capability of our method on the reconstruction of a simple synthetic HRRP. Figure 2 (a) shows the positions of the radar and airplane target in our simulation, where  $X-O-Y-Z$  denotes the coordinate with the airplane target on the center  $O$ ; while Figure 2 (b) shows the scatterer distribution of the airplane target in the coordinate  $X-O-Y$ , where the airplane target is composed of 8 scatterers with all  $Z$  indexes equal to zeros for simplifying our simulation. As shown in Figure 2, the radar line of sight (RLOS) parallels the  $O-Y$  axis. In  $X-O-Y-Z$ , if  $\mathbf{n}_0 = [0, 1, 0]^T$  denotes the RLOS vector and  $\mathbf{s} = [x, y, z]^T$  denotes the scatterer vector, the scatterer location can be calculated via

$$R = \frac{\langle \mathbf{n}_0, \mathbf{s} \rangle}{\|\mathbf{n}_0\|_2} \quad (20)$$

In our simulation, the scattering coefficients and locations of the scatterers are listed in Table II. This synthetic HRRP is much simpler since it disregards the existence of minor scatterers, however it is useful for demonstrating the behavior of the proposed method, and for analyzing the accuracy of the reconstructed scatterer locations based on the known truth. Here  $L = 256$ ,  $\Delta f = \frac{B}{L} = \frac{400}{256} \text{ MHz}$ .

Figures 3 (a) and (b) show the synthetic HRRPs without noise and under SNR=10dB. The denoising results from Figure 3 (b) with  $K = L$  and  $K = 2 \times L$  are shown in Figures 3 (c) and (d) respectively. Since 2 scatterers in the

TABLE II  
SCATTERER LOCATIONS AND SCATTERING COEFFICIENTS OF A SYNTHETIC HRRP, OF WHICH THE  
LOCATIONS ARE MEASURED IN THE  $X - O - Y - Z$  COORDINATE IN FIGURE 2

| scatterer #            | 1      | 2      | 3     | 4      | 5     | 6      | 7     | 8     |
|------------------------|--------|--------|-------|--------|-------|--------|-------|-------|
| X index (m)            | -11.25 | -2.25  | 3.75  | 9.38   | 0     | -15.00 | -3.75 | 0     |
| Y index (m)            | -15.00 | -12.56 | -7.50 | -3.19  | 0     | 3.00   | 7.50  | 11.25 |
| Z index (m)            | 0      | 0      | 0     | 0      | 0     | 0      | 0     | 0     |
| location $R$ (m)       | -15.00 | -12.56 | -7.50 | -3.19  | 0     | 3.00   | 7.50  | 11.25 |
| scattering coefficient | $0.5j$ | 1      | $2j$  | $1-2j$ | $2+j$ | $-2j$  | $-j$  | 0.8   |

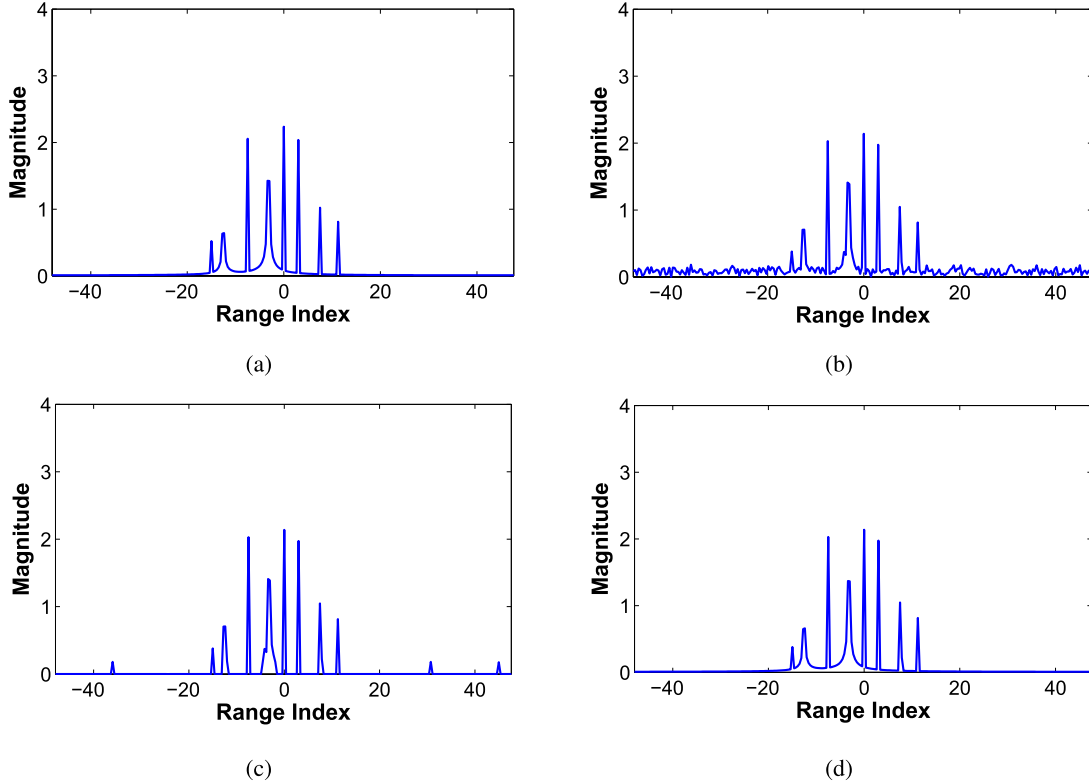


Fig. 3. Denoising example for a synthetic HRRP: (a) the synthetic HRRP without noise; (b) the synthetic HRRP under SNR=10dB; (c) the denoised HRRP from (b) via OMP with  $K = L$ ; and (d) the denoised HRRP from (b) via OMP with  $K = 2 \times L$ .

synthetic HRRP are located in the superresolution grid with  $K = 2 \times L$ , OMP with  $K = 2 \times L$  can accurately extract the 8 scatterers. The detailed location estimation results are given in Table III. There are totally 21 scatterers extracted via OMP with  $K = L$ , of which only the first 10 scatterers with the larger magnitudes are given in Table III.

To quantitatively evaluate the denoising performance, we define the relative HD-based-error as

$$E = \frac{H(\hat{\mathbb{A}}, \bar{\mathbb{A}})}{H(\bar{\mathbb{A}}, \{[0, 0]^T\})} \quad (21)$$

where  $\hat{\mathbb{A}}$  denotes the point set for the denoised HRRP sample,  $\bar{\mathbb{A}}$  denotes the point set for the original HRRP sample without noise,  $H(\cdot, \cdot)$  is calculated according to (17), and the denominator  $H(\bar{\mathbb{A}}, \{[0, 0]^T\})$  is used to normalize the HD-based-error. Here the reason why we use the

HD-based-error rather than the mean squared error (MSE) is that the HD-based-error and the proposed HD-based recognition method (Algorithms 2 and 3) use the same distance metric. As shown in Table IV, OMP with  $K = 2 \times L$  gets the better denoising performance.

2) *Measured Data Reconstruction*: In the following, we show the denoising results on the measured data. Figure 4 gives the three denoising examples for HRRP data under SNR=10dB via OMP with  $K = 2 \times L$ . Comparing between the original HRRP, noisy HRRP and denoised HRRP samples, we can see that the proposed denoising method works well not only in removing the noise component but also in reconstructing the target scattering coefficients.

Figure 5 further quantitatively depicts the denoising performance for all test HRRP samples in our data set via the average relative HD-based-errors. Here the relative HD-based-error

TABLE III  
LOCATION ESTIMATION RESULTS VIA OMP WITH  $K = L$  AND  $K = 2 \times L$  FOR A SYNTHETIC HRRP DATA, WHICH ARE MEASURED IN THE  $X - O - Y - Z$  COORDINATE IN FIGURE 2. HERE WE ONLY GIVE THE RESULTS FOR THE FIRST 10 SCATTERERS WITH THE LARGER MAGNITUDES FROM THE 21 SCATTERERS EXTRACTED VIA OMP WITH  $K = L$

| scatterer #                                                    | 1      | 2      |        | 3     | 4     |       | 5 | 6    | 7    | 8     |
|----------------------------------------------------------------|--------|--------|--------|-------|-------|-------|---|------|------|-------|
| real location $R$<br>( $m$ )                                   | -15.00 | -12.56 |        | -7.50 | -3.19 |       | 0 | 3.00 | 7.50 | 11.25 |
| location estimated via<br>OMP with $K = L$<br>( $m$ )          | -15.00 | -12.75 | -12.38 | -7.50 | -3.38 | -3.00 | 0 | 3.00 | 7.50 | 11.25 |
| location estimated via<br>OMP with $K = 2 \times L$<br>( $m$ ) | -15.00 | -12.56 |        | -7.50 | -3.19 |       | 0 | 3.00 | 7.50 | 11.25 |

TABLE IV  
RELATIVE HD-BASED-ERRORS VIA OMP WITH  $K = L$  AND  $K = 2 \times L$  FOR A SYNTHETIC HRRP DATA

|                                             | relative HD-based-error |
|---------------------------------------------|-------------------------|
| noisy data under SNR=10dB                   | 0.3025                  |
| denoised data via OMP with $K = L$          | 0.0675                  |
| denoised data via OMP with $K = 2 \times L$ | 0.0260                  |

for each test sample is calculated according to (21) with  $\bar{\mathbf{A}}$  denoting the point set for the original HRRP samples under SNR $\approx$ 40dB. We also use five noisy datasets with different adding noise processing to evaluate the robustness of our proposed method. Figure 5 shows the relative HD-based-errors averaged over five runs with different adding noise processing. As shown in Figure 5, the average relative HD-based-errors of the denoised HRRP data are smaller than those of the noisy data, especially under the low SNR conditions. Table V further gives the detailed mean values and corresponding standard deviations of relative HD-based-errors via OMP with  $K = 2 \times L$  over five runs with different adding noise processing. As shown in Table V, the standard deviations of relative HD-based-errors over five runs are very small, therefore, our proposed method is robust to different adding noise processing.

### C. Recognition Performance

In the following classification experiment, we compare the proposed noise-robust scatterer matching recognition method with its non-robust version and the existing statistical recognition method based on Gaussian model [14], [19] without denoising. Figure 6 shows the average recognition rates of our measured dataset versus SNR. The non-robust version of scatterer matching recognition method means the noise power threshold  $\delta$  for a test sample is very small in Algorithm 1, which accords to a high SNR case ( $\approx$ 40dB) and does not equal to its real value. In this way, the noised scatterer information is used in the following HD-based recognition method. As shown in Figure 6, when the SNR $\geq$ 25dB, the influence of noise is not so significant, the four methods yield the similar recognition accuracies, and the Gaussian model is a little better than

other methods only under SNR $\approx$  30dB; when SNR $\leq$ 25dB, the two noise-robust methods outperform the noised Gaussian model and noised HD-based method, and the performance of our superresolution version ( $K = 2 \times L$ ) is a little better than that of the orthogonal version ( $K = L$ ). In the real application, we need a threshold of the recognition rate for the RATR problem to evaluate the noise-robustness performance. According to our experience and some papers based on our measured data from the three real airplanes [4]–[7], [9], [21], recognition rate larger than 80% can satisfy the requirement of the real application. As shown in Figure 6, if we assume the recognition rate threshold is 80%, our method with  $K = 2 \times L$  can work under the test condition of SNR $\geq$ 18dB, while the noised Gaussian model requires SNR $\geq$ 24dB. According to the radar equation, in the real application a 1dB advantage in SNR will bring an increase about 6% in the recognition distance between radar and target. Therefore, the recognition distance of our proposed method is about 1.42 times as long as that of the noised Gaussian model.

To evaluate the robustness of our proposed methods to different adding noise processing, Table VI further gives the detailed mean values and corresponding standard deviations of recognition rates via OMP with  $K = L$  and OMP with  $K = 2 \times L$  over five runs with different adding noise processing. As shown in Table VI, the standard deviations of recognition rates over five runs are very small, therefore, our proposed methods are robust to different adding noise processing.

Since the computation complexity in the training stage can be ignored for an off-line learning (or training) system, it is more important to evaluate the computation complexity in the classification stage. For the HD measure based scatterer matching recognition algorithm, the computation complexity



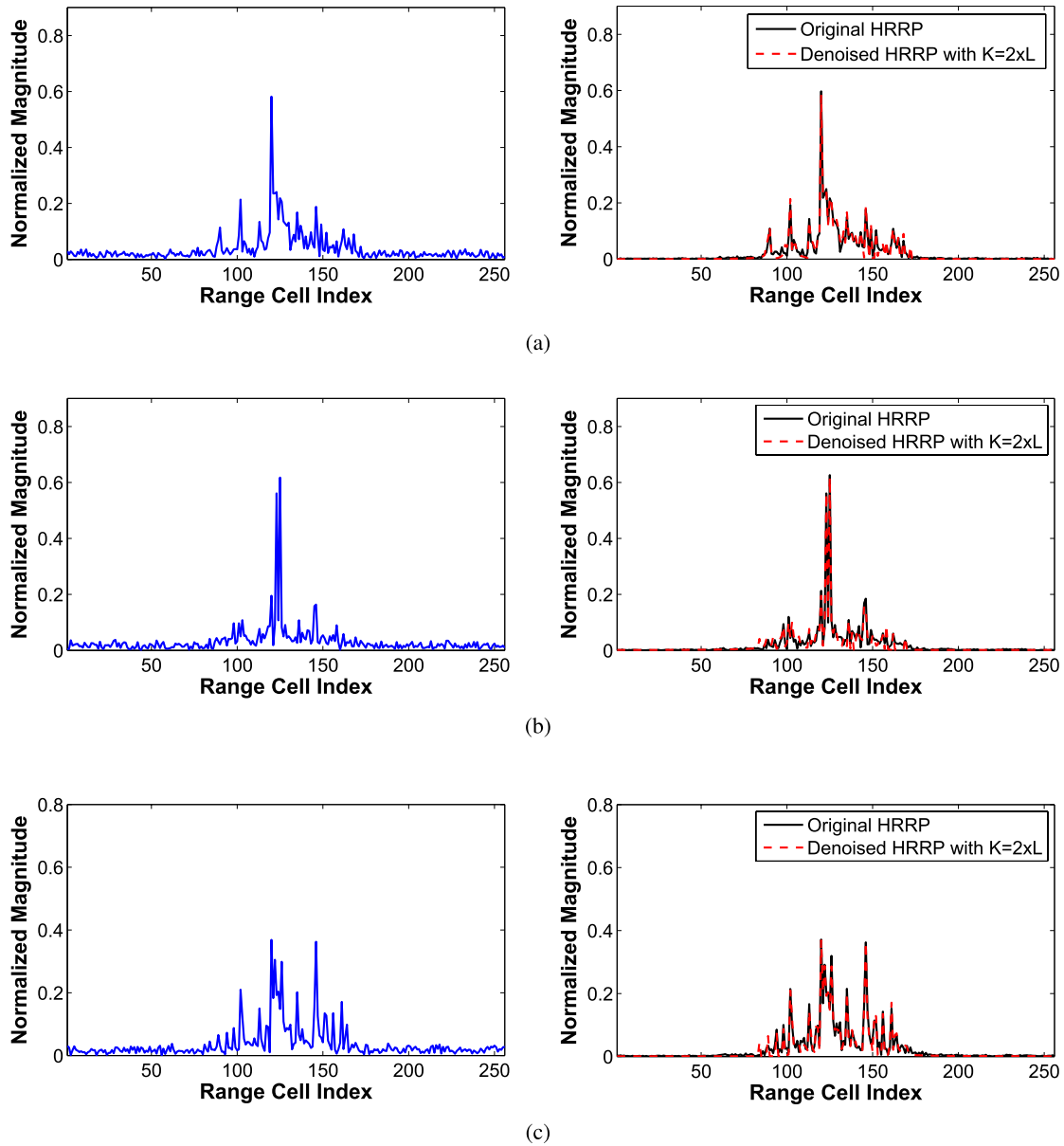


Fig. 4. Denoising examples for three measured HRRP samples of the three airplane targets introduced in Table I. (a) Left column: the noisy HRRP sample of Yark-42 airplane under SNR=10dB; Right column: the original HRRP under high SNR about 40dB (black real line) and the denoised HRRP from that on the left via OMP with  $K = 2 \times L$  (red dot line). (b) Left column: the noisy HRRP sample of Cessna Citation S/II airplane under SNR=10dB; Right column: the original HRRP under high SNR about 40dB (black real line) and the denoised HRRP from that on the left via OMP with  $K = 2 \times L$  (red dot line). (c) Left column: the noisy HRRP sample of An-26 airplane under SNR=10dB; Right column: the original HRRP under high SNR about 40dB (black real line) and the denoised HRRP from that on the left via OMP with  $K = 2 \times L$  (red dot line).

TABLE V

MEAN VALUES AND CORRESPONDING STANDARD DEVIATIONS OF RELATIVE HD-BASED-ERRORS VIA OMP WITH  $K = 2 \times L$  FOR ALL TEST HRRP SAMPLES IN OUR DATASET OVER FIVE RUNS WITH DIFFERENT ADDING NOISE PROCESSING

| SNR (dB)                                                                             | 5               | 10              | 15              | 20              | 25              | 30              | 35              |
|--------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| relative<br>HD-based-errors via<br>OMP with $K = 2 \times L$<br>( $\times 10^{-2}$ ) | $4.00 \pm 0.99$ | $2.78 \pm 0.48$ | $1.90 \pm 0.36$ | $1.63 \pm 0.26$ | $1.33 \pm 0.26$ | $1.34 \pm 0.21$ | $1.30 \pm 0.15$ |

in the classification stage is  $O(10K^* \sum_{g=1}^G \sum_{m=1}^{M_g} K_m^g)$ , where  $K^*$  and  $K_m^g$  respectively denote the numbers of scatterers extracted from the test sample and the frame template in the  $m$ th aspect-frame of the  $g$ th target,  $M_g$  denotes the

number of aspect-frames of the  $g$ th target and  $G$  denotes the number of targets; for the Gaussian model, the computation complexity in the classification stage is  $O(3L \sum_{g=1}^G M_g)$ , where  $L$  denotes the number of frequency components in the

TABLE VI  
MEAN VALUES AND CORRESPONDING STANDARD DEVIATIONS OF RECOGNITION RATES VIA OMP WITH  $K = L$  AND  $K = 2 \times L$  FOR ALL TEST HRRP SAMPLES IN OUR DATASET OVER FIVE RUNS WITH DIFFERENT ADDING NOISE PROCESSING

| SNR (dB)                                            | 5            | 10           | 15           | 20           | 25           | 30           | 35           |
|-----------------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| recognition rates via OMP with $K = L$ (%)          | 64.91 ± 0.33 | 68.12 ± 0.34 | 69.13 ± 0.20 | 76.56 ± 0.49 | 81.33 ± 0.57 | 85.27 ± 0.19 | 87.18 ± 0.25 |
| recognition rates via OMP with $K = 2 \times L$ (%) | 65.59 ± 0.39 | 69.38 ± 0.64 | 71.88 ± 0.46 | 81.80 ± 0.73 | 85.64 ± 0.27 | 85.60 ± 0.01 | 87.18 ± 0.25 |

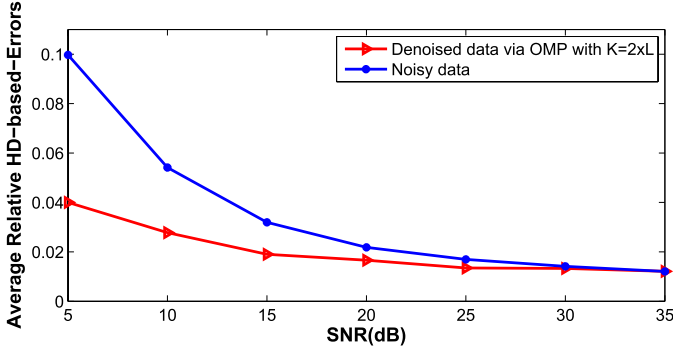


Fig. 5. Variation of the average relative HD-based-errors for all test HRRP samples in our data set with SNR. The blue line denotes the average relative errors between the noisy data and the original data under high SNR about 40dB, while the red line denotes those between the denoised data via OMP with  $K = 2 \times L$  and the original data under high SNR about 40dB. The errors are averaged over five runs with different adding noise processing.

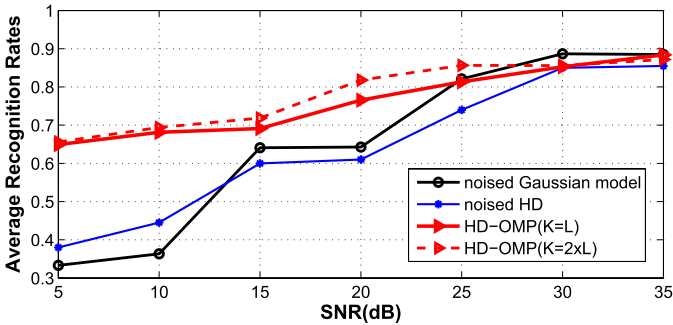


Fig. 6. Variation of the average recognition rates with SNR via statistical recognition based on Gaussian model [14], [19] without denoising, HD-based scatterer matching recognition without denoising for test samples and with test samples denoised via OMP ( $K = L$  and  $K = 2 \times L$ ). The recognition rates are averaged over five runs with different adding noise processing.

test sample. Since  $K * \sum_{g=1}^G \sum_{m=1}^{M_g} K_m^g > L \sum_{g=1}^G M_g$ , the computation complexity of the HD measure based scatterer matching recognition algorithm is larger than that of the Gaussian model in the classification stage.

## V. CONCLUSION

A noise-robust scatterer matching recognition method is proposed for radar HRRP data in this paper. The target dominant scatterers in noisy HRRP data are first extracted via the OMP algorithm, and then the magnitudes and locations of extracted scatterers are used as the feature patterns in the scatterer matching recognition algorithm based on HD. Experimental results on the synthetic and measured data show that the proposed method can obtain good recognition and

denoising performances under the relatively low test SNR condition. In the real application, the SNR advantage can effectively extend the recognition distance between the target and radar.

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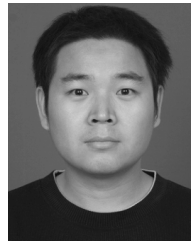
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